

TECHSTART ENTERPRISE NETWORK

Enterprise Network Design and Implementation

Cisco Packet Tracer Final Project Report

COMP 3115 - Practical Network Principles

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Executive Summary

This report documents the design, implementation, security configuration, and verification of the TechStart company network in Cisco Packet Tracer. The network consists of three independent departments - Development, Sales, and Human Resources - connected to a central core router and a server room. The implementation uses variable-length subnetting, DHCP for employee PCs, static addressing for infrastructure devices, OSPF dynamic routing, DNS, HTTP, FTP, SSH-based device management, and extended access control lists for the optional bonus requirements.

Implementation Status

All mandatory network requirements were completed and verified. Interdepartment communication, server access, DNS name resolution, web access, FTP authentication, OSPF route exchange, DHCP allocation, and the two bonus ACL restrictions were successfully tested.

1. Project Overview and Objectives

TechStart is a technology company with three departments located on separate floors. Each department requires an independent LAN with its own router and switch. All department routers connect to CORE-R1 through point-to-point /30 WAN links. CORE-R1 also provides access to a central server room containing DNS, Web, and FTP servers.

- Design seven logical networks from the parent block 10.0.0.0/8.
- Provide capacity for up to 50 hosts in each department LAN.
- Use static IP addresses for router interfaces, switch management interfaces, and servers.
- Use DHCP only for the nine employee PCs.
- Provide secure remote device administration using SSH version 2.
- Enable centralized DNS, HTTP, and FTP services.
- Provide full routing between departments and the server room using OSPF.
- Apply optional ACL restrictions without interrupting permitted traffic.

2. Network Topology and Design

The topology follows a hierarchical star design. CORE-R1 is the central routing device. Each department contains one router, one Cisco 2960 switch, and three PCs. The server room contains SW-Servers and three statically addressed servers. The implemented Packet Tracer topology uses serial WAN interfaces while preserving the required point-to-point /30 addressing and logical connectivity.

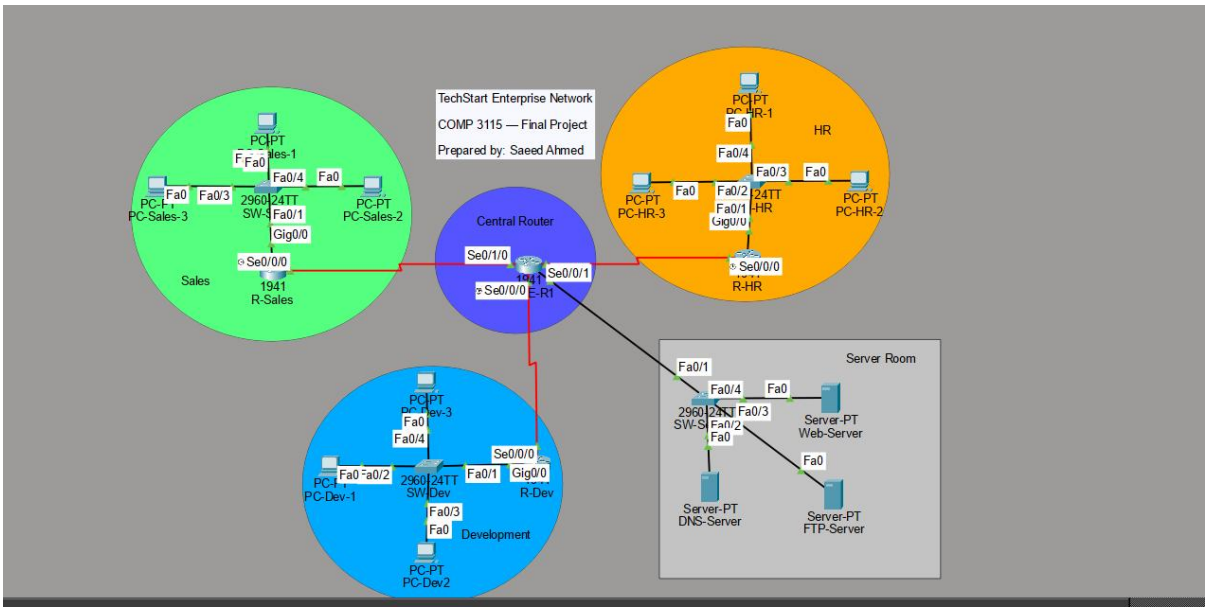


Figure 1. Final TechStart network topology in Cisco Packet Tracer.

3. IP Addressing and Subnetting Plan

Variable-length subnet masks were used to allocate address space efficiently. A /26 subnet provides 62 usable addresses for each department, a /28 subnet provides 14 usable addresses for the server room, and each /30 WAN subnet provides exactly two usable addresses.

Network	Network/CIDR	Mask	First Host	Last Host	Broadcast	Hosts
LAN Dev	10.0.0.0/26	255.255.255.192	10.0.0.1	10.0.0.62	10.0.0.63	62
LAN Sales	10.0.0.64/26	255.255.255.192	10.0.0.65	10.0.0.126	10.0.0.127	62
LAN HR	10.0.0.128/26	255.255.255.192	10.0.0.129	10.0.0.190	10.0.0.191	62
Server Room	10.0.0.192/28	255.255.255.240	10.0.0.193	10.0.0.206	10.0.0.207	14
WAN Dev-Core	10.0.0.208/30	255.255.255.252	10.0.0.209	10.0.0.210	10.0.0.211	2
WAN Sales-Core	10.0.0.212/30	255.255.255.252	10.0.0.213	10.0.0.214	10.0.0.215	2
WAN HR-Core	10.0.0.216/30	255.255.255.252	10.0.0.217	10.0.0.218	10.0.0.219	2

3.1 Device Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R-Dev	G0/0 (LAN)	10.0.0.1	255.255.255.192	-
R-Dev	S0/0/0 (WAN)	10.0.0.209	255.255.255.252	-
R-Sales	G0/0 (LAN)	10.0.0.65	255.255.255.192	-
R-Sales	S0/0/0 (WAN)	10.0.0.213	255.255.255.252	-
R-HR	G0/0 (LAN)	10.0.0.129	255.255.255.192	-
R-HR	S0/0/0 (WAN)	10.0.0.217	255.255.255.252	-
CORE-R1	S0/0/0 (WAN-Dev)	10.0.0.210	255.255.255.252	-
CORE-R1	S0/1/0 (WAN-Sales)	10.0.0.214	255.255.255.252	-
CORE-R1	S0/0/1 (WAN-HR)	10.0.0.218	255.255.255.252	-
CORE-R1	G0/0 (Server LAN)	10.0.0.193	255.255.255.240	-
SW-Dev	VLAN 1	10.0.0.2	255.255.255.192	10.0.0.1
SW-Sales	VLAN 1	10.0.0.66	255.255.255.192	10.0.0.65
SW-HR	VLAN 1	10.0.0.130	255.255.255.192	10.0.0.129
SW-Servers	VLAN 1	10.0.0.194	255.255.255.240	10.0.0.193
DNS Server	NIC	10.0.0.195	255.255.255.240	10.0.0.193
Web Server	NIC	10.0.0.196	255.255.255.240	10.0.0.193
FTP Server	NIC	10.0.0.197	255.255.255.240	10.0.0.193

3.2 DHCP Pool Design

Department	Pool Name	Network	Excluded Addresses	Default Router	DNS Server
Development	DEV-POOL	10.0.0.0/26	10.0.0.1 - 10.0.0.10	10.0.0.1	10.0.0.195
Sales	SALES-POOL	10.0.0.64/26	10.0.0.65 - 10.0.0.74	10.0.0.65	10.0.0.195
HR	HR-POOL	10.0.0.128/26	10.0.0.129 - 10.0.0.138	10.0.0.129	10.0.0.195

4. Device Configuration and Security

All routers and switches were configured with consistent naming, management security, and saved startup configurations. Router LAN and WAN interfaces were assigned static addresses, switch VLAN 1 interfaces were assigned static management addresses, and all active interfaces were enabled.

Security Item	Configured Value
Enable secret	TechStart@2025
Console password	Console@2025
MOTD banner	Authorized Access Only - TechStart Network
Domain name	techstart.local
Local SSH username	netadmin
Local SSH secret	Admin@2025
RSA key size	1024 bits

Security Item	Configured Value
SSH version	Version 2
VTY access	login local; transport input ssh
Configuration saving	copy running-config startup-config on all routers and switches

Address Assignment Policy

All router interfaces, switch SVIs, and servers use static addresses. Only department PCs obtain addresses dynamically through DHCP. This prevents infrastructure addresses from changing and keeps management and service access predictable.

5. Network Services

5.1 DNS Service

DNS Record	Type	Address
www.techstart.local	A	10.0.0.196
ftp.techstart.local	A	10.0.0.197
dns.techstart.local	A	10.0.0.195

5.2 HTTP Service

The Web Server uses the static address 10.0.0.196. HTTP was enabled and the home page was customized to display the required message: "Welcome to TechStart Company Network." The page also presents department networks, service status, and routing information.

5.3 FTP Service

Server	Address	Username	Password	Permissions
FTP Server	10.0.0.197	techuser	Tech@123	Read / Write

6. Routing Design - OSPF

OSPF process 1 in area 0 was selected as the routing protocol. OSPF automatically exchanges routes between the department routers and CORE-R1. It was chosen instead of static routing because it is easier to manage, adapts automatically to topology changes, and scales better when new networks are added.

- R-Dev advertises 10.0.0.0/26 and 10.0.0.208/30.
- R-Sales advertises 10.0.0.64/26 and 10.0.0.212/30.
- R-HR advertises 10.0.0.128/26 and 10.0.0.216/30.
- CORE-R1 advertises the three WAN networks and 10.0.0.192/28.
- All three OSPF adjacencies on CORE-R1 reached the FULL state.

Routing Verification Result

CORE-R1 learned the three department LANs as OSPF routes, while each department router learned the other departments and the server room. No default route was required because the project does not include an Internet connection.

7. Verification and Testing

The following evidence confirms that routing, DHCP, interdepartment communication, server access, DNS name resolution, HTTP, and FTP are functioning correctly.

7.1 Routing Table on CORE-R1

CORE-R1

Physical Config CLI Attributes

```
L       10.0.0.210/32 is directly connected, Serial0/0/0
C       10.0.0.212/30 is directly connected, Serial0/1/0
L       10.0.0.214/32 is directly connected, Serial0/1/0
C       10.0.0.216/30 is directly connected, Serial0/0/1
L       10.0.0.218/32 is directly connected, Serial0/0/1

CORE-R1#
CORE-R1#
CORE-R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
CORE-R1#
CORE-R1#
CORE-R1#
CORE-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 11 subnets, 4 masks
O       10.0.0.0/26 [110/65] via 10.0.0.209, 00:10:30, Serial0/0/0
O       10.0.0.64/26 [110/65] via 10.0.0.213, 00:10:30, Serial0/1/0
O       10.0.0.128/26 [110/65] via 10.0.0.217, 00:10:30, Serial0/0/1
C       10.0.0.192/28 is directly connected, GigabitEthernet0/0
L       10.0.0.193/32 is directly connected, GigabitEthernet0/0
C       10.0.0.208/30 is directly connected, Serial0/0/0
L       10.0.0.210/32 is directly connected, Serial0/0/0
C       10.0.0.212/30 is directly connected, Serial0/1/0
L       10.0.0.214/32 is directly connected, Serial0/1/0
C       10.0.0.216/30 is directly connected, Serial0/0/1
L       10.0.0.218/32 is directly connected, Serial0/0/1

CORE-R1#
```

Figure 2. CORE-R1 routing table showing all three department LANs as OSPF routes.

7.2 Routing Table on R-Dev

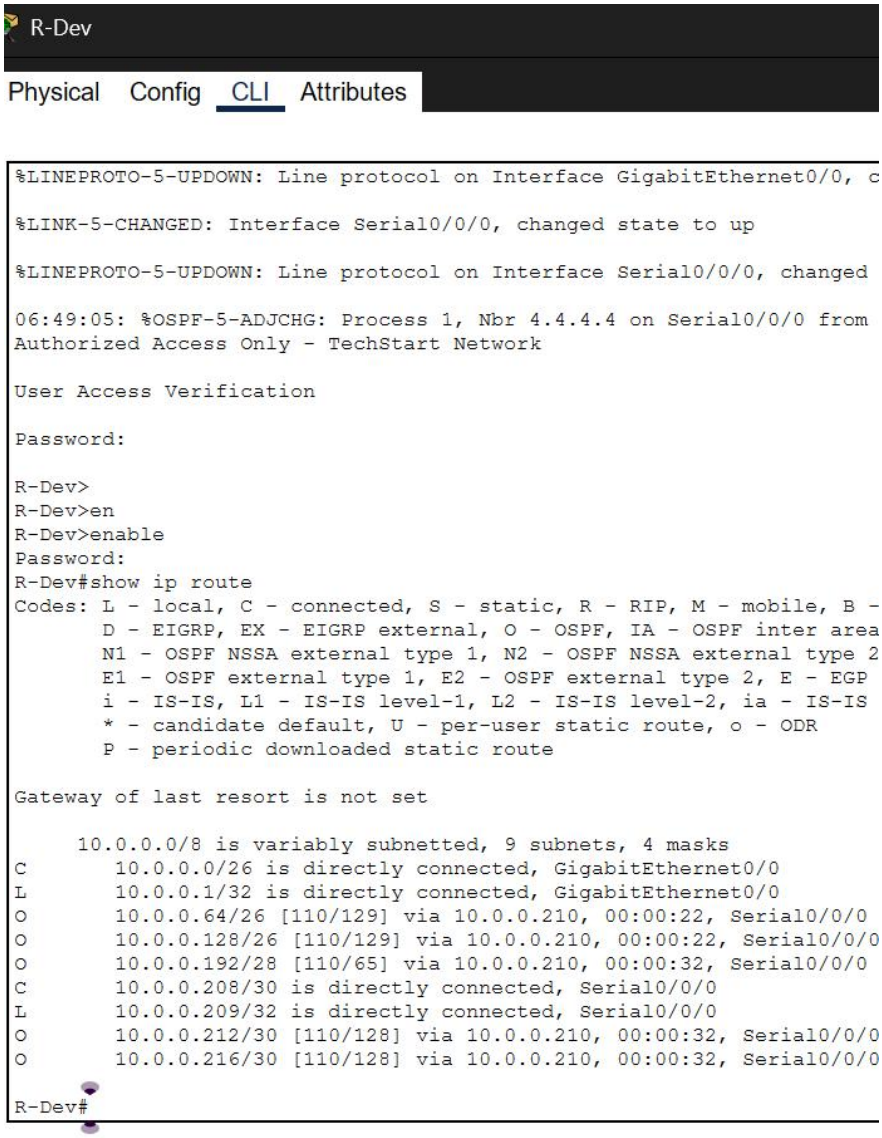


Figure 3. R-Dev routing table showing routes to Sales, HR, and the server room.

7.3 DHCP Address Allocation

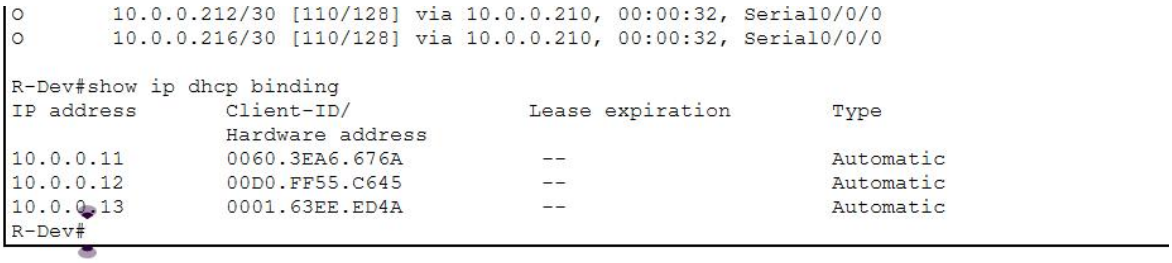


Figure 4. DHCP bindings on R-Dev for the three Development PCs.

7.4 Interdepartment Connectivity

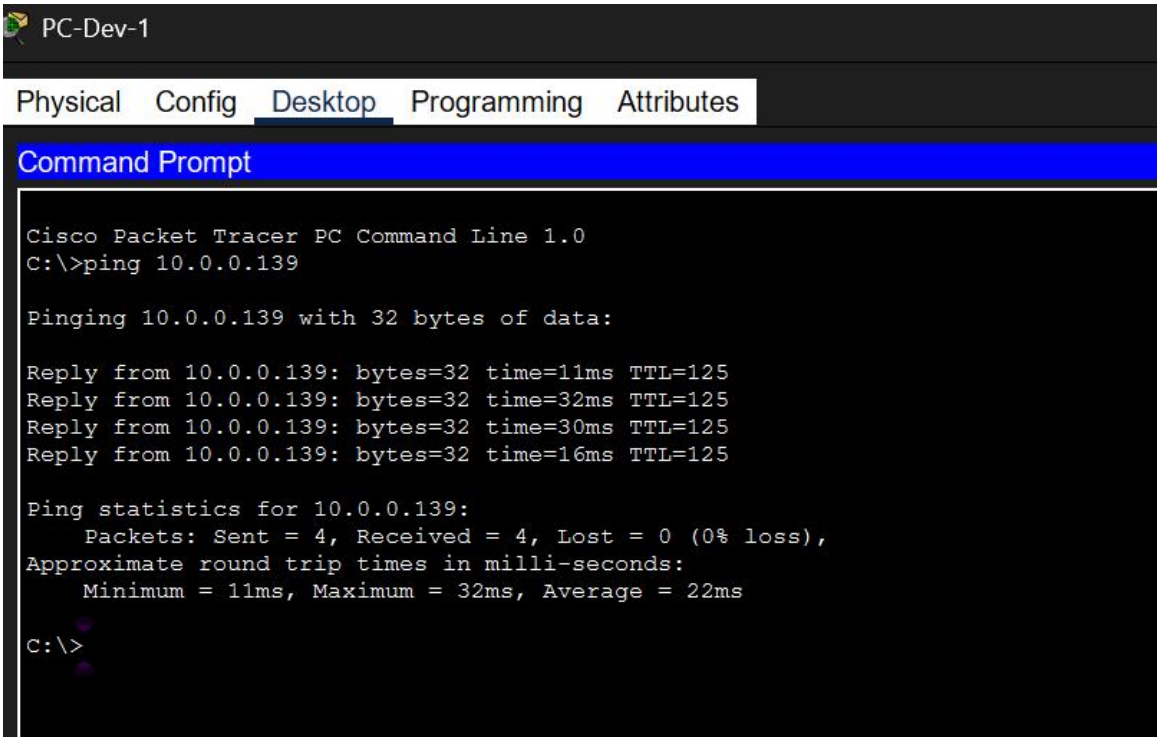


Figure 5. Successful ping from PC-Dev-1 to PC-HR-1 (10.0.0.139), with 0% packet loss.

7.5 Department-to-Server Connectivity

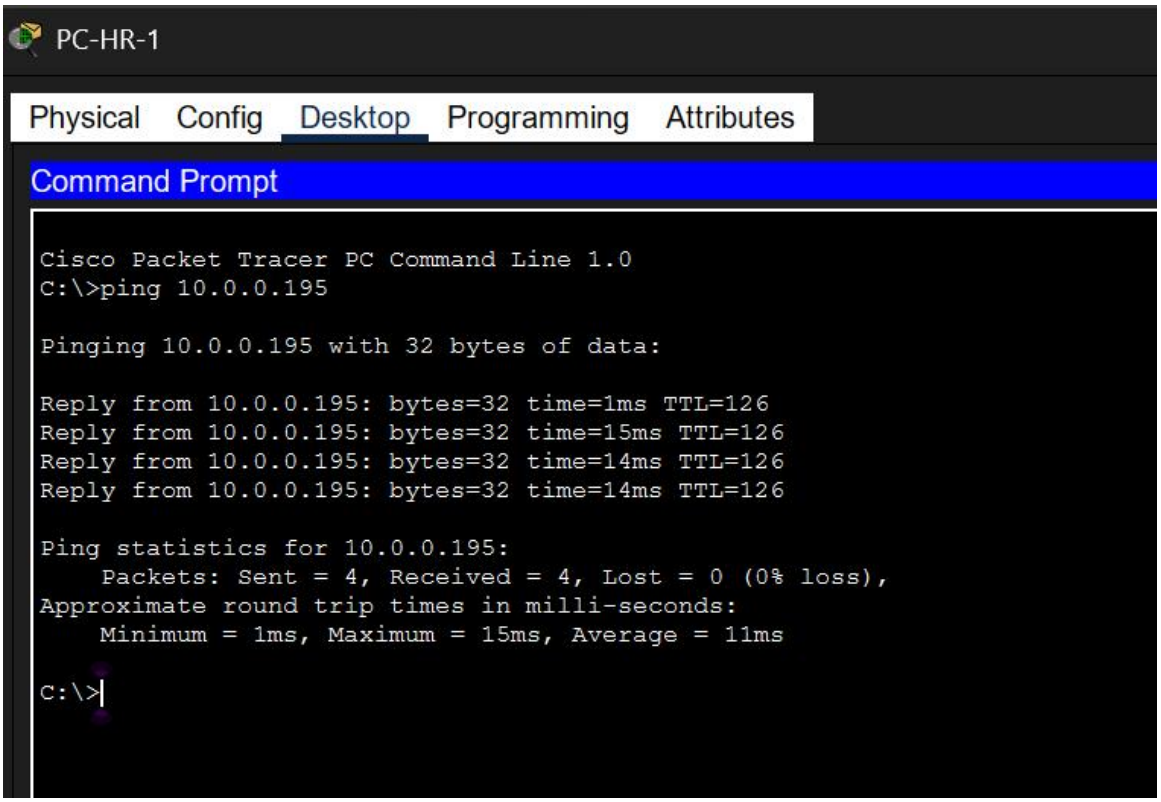


Figure 6. Successful ping from PC-HR-1 to the DNS Server (10.0.0.195).

7.6 Web Service Verification

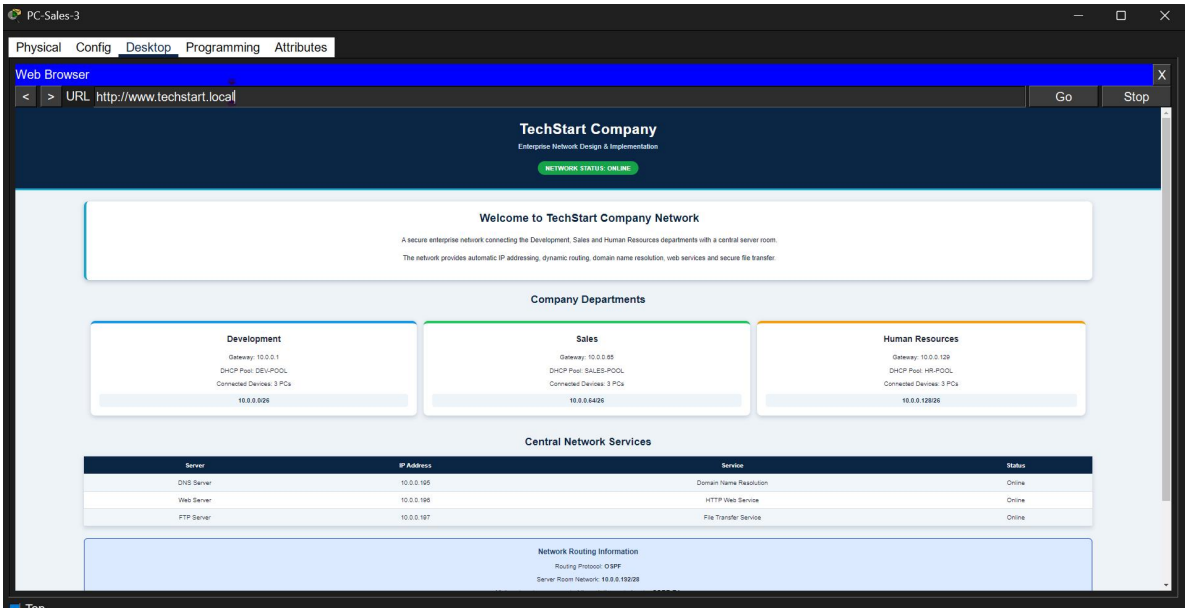


Figure 7. Customized TechStart website opened through http://www.techstart.local from PC-Sales-3.

7.7 FTP Service Verification

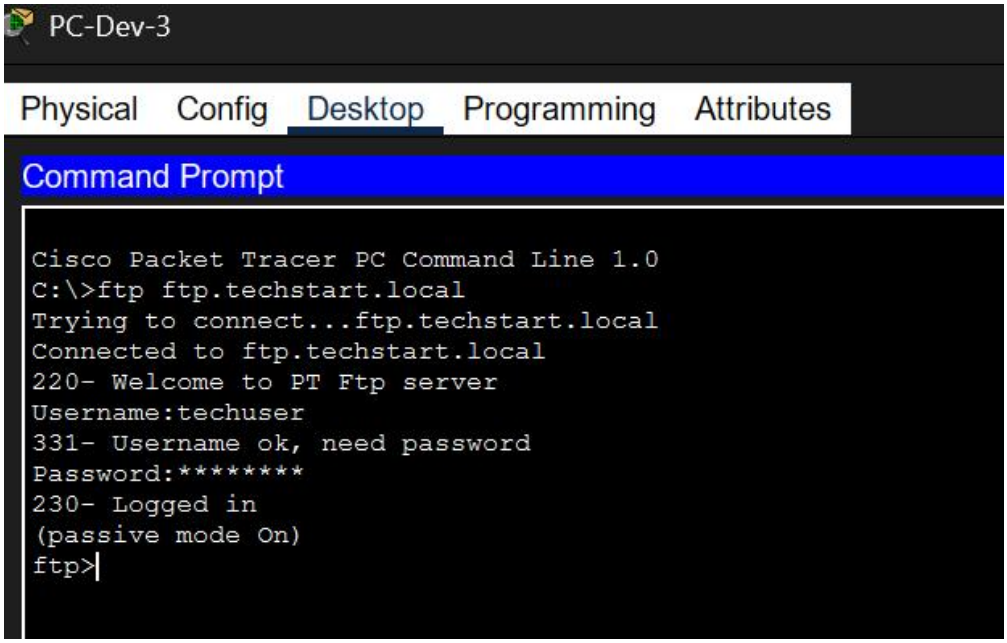


Figure 8. Successful FTP login using the techuser account, confirmed by the 230 Logged in message.

8. Bonus - Access Control Lists

Two extended ACLs were configured close to their source networks and applied inbound on the department LAN interfaces. Each ACL includes an explicit deny statement followed by permit ip any any so that only the required traffic is blocked.

ACL	Router / Interface	Restriction	Permit Rule
HR_FTP_BLOCK	R-HR G0/0 inbound	Deny 10.0.0.128/26 to host 10.0.0.197	Permit all other IP traffic
SALES_DEV_BLOCK	R-Sales G0/0 inbound	Deny 10.0.0.64/26 to 10.0.0.0/26	Permit all other IP traffic

8.1 HR Restriction Test

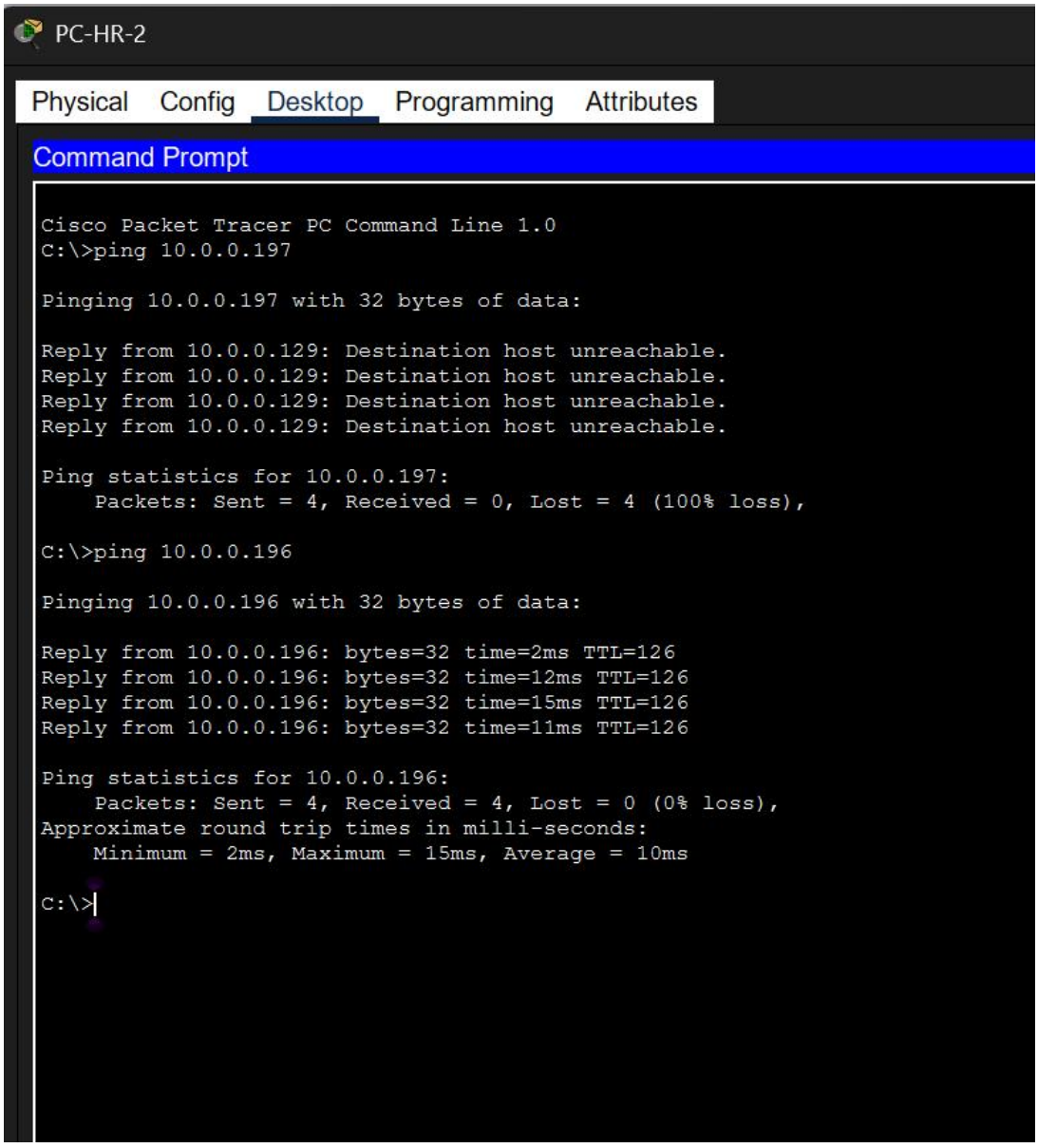


Figure 9. HR is blocked from the FTP Server (10.0.0.197) but can reach the Web Server (10.0.0.196).

8.2 Sales Restriction Test

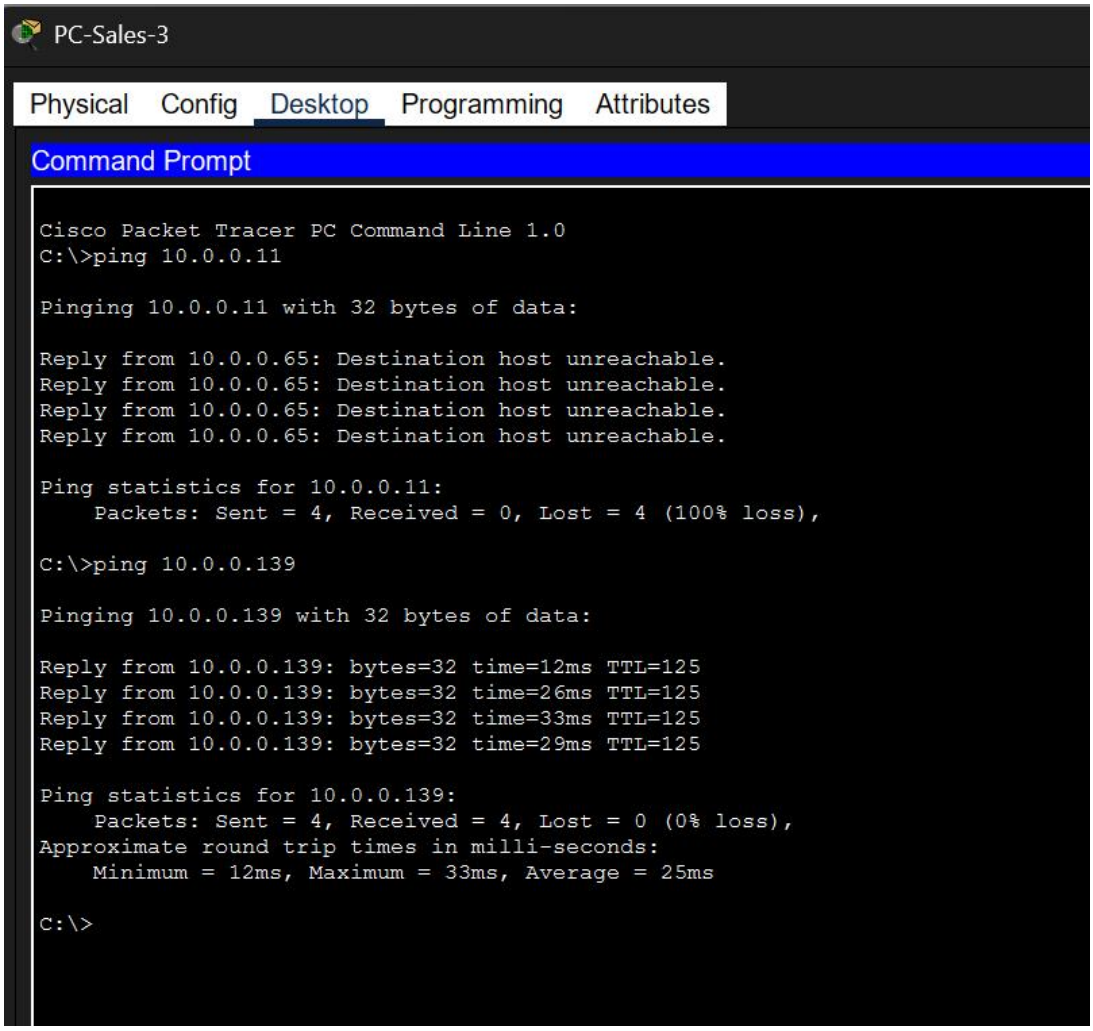


Figure 10. Sales is blocked from the Development LAN but can communicate with the HR LAN.

9. Conclusion

The TechStart enterprise network was successfully designed and implemented in Cisco Packet Tracer. The final solution provides separated departmental networks, efficient subnet allocation, automatic client addressing, secure infrastructure management, centralized services, and dynamic route exchange. Testing confirmed successful communication between permitted networks and services. The optional ACL controls also operated correctly by blocking HR access to the FTP Server and Sales access to the Development LAN while allowing other traffic. The completed design satisfies the mandatory project requirements and the bonus security requirements.

Final Result

All mandatory deliverables were completed: subnetting, addressing, DHCP, secure device configuration, DNS, HTTP, FTP, OSPF routing, verification screenshots, and the optional ACL bonus.